$$\left[\begin{array}{ccc|ccc} 1 & \frac{-1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{3}{4} & \frac{-1}{4} & 1 & 0 \\ 0 & \frac{15}{2} & \frac{-5}{4} & \frac{-9}{4} & 0 & 1 \end{array}\right] \begin{array}{l} A_3 = \\ B_3 = \\ C_3 = \end{array}$$